FAST National University of Computer and Emerging Sciences

Department of Computer Science

PROJECT BRIEF

AI-Powered Multi-Modal Content Generation

Date: March 5, 2026 **Version:** v1.0

Stakeholders and Roles

Project Supervisor	Mr. Muhammad Aamir Gulzar
Lead Developer	Amin Fahim
Developer	Sarim Farooq
Developer	Musab Waseem

This is a Final Year Project (FYP). The supervisor acts as both project sponsor and academic reviewer. The three developers share project management responsibilities jointly.

Background and Context

Demand for video content is growing rapidly across social media and educational platforms, yet professional-quality production remains out of reach for most individual creators due to cost, equipment, and technical complexity. This gap is especially pronounced for Urdu-speaking creators, where tools that understand Urdu's linguistic patterns, phonetics, and cultural context are virtually nonexistent.

This Final Year Project addresses that gap by building an end-to-end, AI-powered web platform that takes a creator from a topic idea to a finished lip-synced video in four automated stages: **AI script generation**, **text-to-speech synthesis (English and Urdu)**, **zero-shot voice cloning**, and **lip synchronization**. No prior technical expertise is required; a user uploads a short reference audio clip and a frontal-face video template, provides a topic, and the platform produces a lip-synced video in their own voice.

The system is implemented as a Next.js web application backed by a Node.js/Express API, a Python-based AI microservices layer, and a Supabase (PostgreSQL) database, with Clerk handling authentication and session management.

Goals and Success Metrics

	AI-assisted script creation		
	Natural English TTS output		
	Natural Urdu TTS output		
	Zero-shot voice cloning		Clone user voice fro
	Lip synchronization		
	End-to-end integrated pipeline		

Scope and Constraints

In Scope

- Web application (Next.js 14, TypeScript) with Clerk authentication, protected dashboard, and user management
- AI script generation and iterative refinement in English and Urdu using LLMs via OpenRouter (Claude, GPT-4, Gemini), with version history and draft saving
- Six-stage content creation workflow: language selection → script method → script generation/refinement → script review → TTS synthesis → lip synchronization
- English TTS via a Coqui TTS FastAPI microservice
- Urdu TTS via a two-stage pipeline: Parler-TTS (Indic) for base phonetic synthesis, followed by OpenVoice V2 zero-shot tone-color transfer
- Lip synchronization using Wav2Lip on user-uploaded frontal-face video templates
- Node.js/Express backend with Supabase (PostgreSQL), Clerk JWT-secured API, and RLS-enforced data isolation
- Responsive dark-theme UI with real-time progress tracking for long-running AI jobs

Out of Scope

- Mobile applications (iOS or Android)
- Real-time or streaming video generation
- Multi-speaker or dialogue-based videos
- Languages other than English and Urdu
- Per-user custom model training (zero-shot approach eliminates this)
- Post-production video editing beyond lip synchronization

Constraints

- Three-person development team with a fixed FYP academic deadline
- GPU inference for Wav2Lip and TTS models limited to local and free-tier compute; processing latency will vary by hardware availability

- Urdu output quality bounded by the current capability of the open-source Parler-TTS Indic model
- No allocated budget for paid cloud GPU instances; all third-party services use free or academic tiers

Key Deliverables

	Authenticated Web Application	
	Script Generation & Refinement Module	
	English TTS Service	
	Urdu Voice Cloning Pipeline	
	Lip Synchronization Service	
	End-to-End Integrated Workflow	
	FYP Report & Documentation	

Timeline and Milestones

	Phase 1: Foundation & Core Setup		S
	Phase 2: Voice Technology & Content Processing		
	Phase 3: Video Pipeline & Lip Synchronization		I
	Phase 4: Integration & Final Polish		

Resources and Dependencies

Team

- **Amin Fahim** — Lead developer; full-stack and AI pipeline integration
- **Sarim Farooq** — Backend engineer; API, database, video pipeline orchestration
- **Musab Waseem** — Frontend engineer; UI/UX, component library, TTS microservice
- **Mr. Muhammad Aamir Gulzar** — Supervisor; academic guidance, milestone reviews, final sign-off

Budget: Academic FYP with no formal budget. Relies on free-tier Supabase, Vercel/localhost deployment, and locally available GPU compute for AI model inference.

External Dependencies

	OpenRouter API	
	Clerk	
	Supabase	
	Parler-TTS (Indic)	
	OpenVoice V2 (MyShell)	
	Coqui TTS	
	Wav2Lip	
	Whisper (OpenAI)	

Risks and Assumptions

	High inference latency on limited GPU	
	Urdu TTS naturalness bounded by Parler-TTS Indic model	
	OpenRouter API rate limits or key expiry	
	Wav2Lip quality degrades on non-frontal or low-resolution video	
	LLM produces unnatural or overly formal Urdu scripts	
	Zero-shot cloning requires $\geq$15 s of clean reference audio	
	All four phases completable within the FYP deadline	

Supporting Documents

- **System Architecture:** `architecture/SYSTEM_ARCHITECTURE.md`
- **Design System:** `architecture/DESIGN_SYSTEM.md`
- **Backend Implementation Guide:** `server/CONTENT_BACKEND_README.md`
- **Urdu Voice Cloning Logic:** `urdu_voice_clone_logic.txt`
- **Database Schema:** `server/CREATE_SCRIPTS_TABLE.sql`
- **Authentication Setup:** `CLERK_SETUP.md`
- **Script Generation Analysis:** `SCRIPT_GENERATION_REFINEMENT_ANALYSIS.md`

Approvals

	Mr. Muhammad Aamir Gulzar		Supervisor / Sponsor	
	Amin Fahim		Lead Developer	
	Sarim Farooq		Developer	
	Musab Waseem		Developer	

Version History: v1.0 — Initial brief created March 5, 2026. This document is the reference for scope, goals, and ownership decisions throughout the remaining development phases. Any scope changes after Phase 3 kickoff must be reviewed and approved by the supervisor before being reflected here.